

Multi-Source Transfer Learning for Facial Pigmented Lesion Classification in Data-Limited Latin American Dermatology Contexts

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Abstract. Facial pigmented lesions present substantial diagnostic complexity because benign, premalignant, and malignant entities may share overlapping dermoscopic patterns. This study proposes a transfer-learning framework for the seven-class classification of facial pigmented lesions using a pre-trained foundation model. **Methods:** To address target-domain data scarcity, the original MEDICON dataset was expanded by harvesting external public datasets, including HAM10000, PAD-UFES-20, ISIC 2019, DERM12345, BCN20000, and ISIC_ARCHIVE. These datasets were mapped to the target 7-class taxonomy and directly merged to create a unified training pool of 11,020 images. The proposed vision-only pipeline extracts 6144-dimensional embeddings using the Google Derm Foundation model [1], followed by a lightweight Multi-Layer Perceptron classifier. **Results:** Preliminary validation showed moderate discriminative performance, with Macro-AUC of 0.6838, a Brier Score of 0.1079, and an Accuracy of 0.3886. However, class-balanced classification remains limited, reflected by an Average Recall of 0.2904 and a Macro-F1 of 0.3012. **Conclusions:** Foundation model embeddings can support facial pigmented lesion classification by effectively extracting visual features without computationally expensive pretraining.

Keywords: Facial pigmented lesions · Transfer learning · Dermatological artificial intelligence

1. Introduction

Skin cancer and pigmented skin lesions remain a major diagnostic challenge in dermatology, particularly when timely specialist evaluation and dermoscopic assessment are not uniformly available across healthcare systems.[2] In Latin America, this problem is especially relevant because healthcare access, specialist availability, cancer registry coverage, and digital infrastructure vary substantially across countries and between urban and non-urban settings.[3] For artificial intelligence (AI)-assisted dermatology, these constraints create a dual challenge: models must not only achieve high predictive performance, but also demonstrate robustness under data scarcity, population heterogeneity, and clinically realistic acquisition conditions.[4]

Facial pigmented lesions are particularly difficult to classify because benign, premalignant, and malignant entities may share overlapping visual patterns. Lesions such as atypical nevus, lentigo maligna, lentigo maligna melanoma, pigmented actinic keratosis, seborrheic keratosis, seborrheic-lichenoid keratosis, and solar lentigo can present with subtle differences in pigmentation, borders, texture, and anatomical distribution.[5] These diagnostic challenges are amplified in sun-exposed facial areas, where chronic photodamage and age-related changes may obscure visual cues.[6] Therefore, classification systems for facial pigmented lesions should not rely exclusively on global accuracy, but should also be evaluated using class-balanced metrics and calibration-aware measures.

Deep learning has shown strong potential for skin lesion classification, largely due to the visual nature of dermoscopic diagnosis and the availability of large public datasets. Datasets such as HAM10000, ISIC challenge datasets, BCN20000, and DERM12345 have enabled the development of models trained on thousands of dermoscopic images across multiple diagnostic categories [7]. HAM10000 contains 10,015 dermoscopic images from common pigmented lesions [8], while BCN20000 was designed to represent more unconstrained “in-the-wild” dermoscopic conditions, including difficult anatomical sites and challenging lesion presentations [9]. More recently, DERM12345 provided a large multisource dermoscopic dataset with fine-grained subclass labels, further supporting representation learning for dermatological AI [10].

However, the direct translation of models trained on large public dermatology datasets to new clinical populations remains limited by domain shift, label heterogeneity, and incomplete representation of diverse populations. A scoping review in JAMA Dermatology found that many dermatology AI datasets lacked adequate reporting of patient skin tone, race, ethnicity, and label quality, raising concerns about bias, reproducibility, and clinical generalizability [11]. Similarly, recent systematic evidence suggests that although AI systems can achieve promising skin-lesion classification performance, their fairness, external validity, and reliability across diverse clinical settings remain uncertain [12]. These concerns are highly relevant for Latin America, where local population structure, ancestry, skin phototype distribution, imaging workflows, and health-system constraints may differ from the settings in which many public dermatology datasets were originally collected.

The scarcity of locally representative dermatology datasets is a key limitation for AI development in Latin American contexts. In this regard, PAD-UFES-20 is particularly important because it was collected in Brazil and includes clinical skin lesion images acquired with smartphone devices [13]. Unlike most dermoscopy-centered datasets, PAD-UFES-20 reflects a more pragmatic

image acquisition scenario, making it relevant for studying AI-assisted dermatology in resource-constrained or non-specialized environments. Nevertheless, PAD-UFES-20 differs from dermoscopic datasets in imaging modality, diagnostic taxonomy, and acquisition conditions, which reinforces the need to carefully map and harmonize disparate data sources into a unified taxonomy to effectively expand the training pool.

From a Latin American perspective, this work frames AI-assisted dermatology as a potential decision-support and triage tool rather than an autonomous diagnostic replacement. This distinction is essential because poor recall for rare, malignant, or premalignant categories could worsen clinical risk if deployed without adequate validation. Accordingly, our evaluation emphasizes macro-averaged performance, average recall, macro-F1, probability calibration, and overall accuracy. This metric selection is intended to capture not only whether the model performs well on frequent classes, but also whether it behaves reliably across clinically relevant categories under data-scarce conditions.

The main contributions of this study are threefold. First, we present a vision-only transfer learning framework that leverages a pre-trained foundation model [1] and a harmonized, multi-source dataset of 11,020 images to overcome target-domain data scarcity. Second, we analyze model performance using class-balanced and calibration-aware metrics, highlighting the gap between ranking ability and clinically meaningful class-level prediction. Third, we discuss the relevance and limitations of such systems for Latin American healthcare contexts, where local validation, population diversity, and equitable access to dermatological expertise remain central challenges.

2. Problem Statement

Pigmented skin lesions represent a clinically relevant problem in Latin America because they include a broad spectrum of benign, premalignant, and malignant conditions with partially overlapping visual characteristics. Although many pigmented lesions are benign, some may correspond to melanoma or to lesions associated with malignant transformation risk, while others, such as pigmented actinic keratosis or lentigo maligna, require timely clinical evaluation or histopathological confirmation. This problem becomes particularly important in chronically sun-exposed anatomical regions, such as the face and neck, where cumulative ultraviolet radiation, photoaging, and background pigmentation changes may complicate lesion assessment.

The epidemiological burden of melanoma in Latin America is generally lower than in regions such as Oceania, North America, and Northern Europe; however, this does not mean that the problem is clinically negligible. Global analyses have shown that melanoma incidence and mortality vary considerably across regions, and that Latin America reports lower rates than high-incidence regions while still facing important diagnostic and public-health challenges [14]. In addition, studies on cancer burden in Latin America and the Caribbean highlight persistent inequalities across countries and the need for cancer control strategies adapted to regional contexts, particularly because population-based data quality and access to healthcare services remain heterogeneous [15].

A central limitation in the region is the incomplete characterization of skin cancer epidemiology. In several Latin American countries, population-based cancer registry coverage is limited or uneven, which may affect the accuracy of melanoma and other skin cancer incidence and mortality estimates. Population-based evidence from Colombia, for example, indicates that epidemiological research on cutaneous melanoma in Latin America remains scarce [16] and that the disease may present regional patterns different from those observed in other populations, including a relatively higher proportion of acral lentiginous melanoma in Latin American cohorts. This is relevant because incomplete epidemiological surveillance can delay public-health responses, reduce awareness campaigns, and limit the development of locally validated diagnostic tools.

The clinical problem is not limited to melanoma alone. Latin America includes populations exposed to high levels of solar radiation throughout much of the year, particularly in tropical, subtropical, coastal, and high-altitude regions. Chronic photodamage contributes to the development of actinic keratoses and other sun-related lesions, some of which may appear pigmented and clinically mimic melanoma or lentigo maligna. A Latin American consensus on actinic damage emphasized the relevance of actinic keratosis and photodamage management in the region, while a meteorological study reported the feasibility of daylight-mediated therapies in Central and South America due to favorable solar radiation conditions [17, 18]. Although these studies mainly focus on treatment and therapeutic feasibility, they support the regional importance of sun-related skin damage.

Facial pigmented lesions are particularly problematic because several entities may resemble one another clinically and dermoscopically. Solar lentigo, pigmented actinic keratosis, seborrheic keratosis, lentigo maligna, and lentigo maligna melanoma may

coexist in older patients with chronic sun exposure and facial photodamage. This visual overlap increases diagnostic uncertainty and may require specialist evaluation, dermoscopy, sequential monitoring, or biopsy. In Latin American contexts, where access to dermatological services may be unequal, this diagnostic complexity may contribute both to delayed diagnoses and unnecessary referrals, depending on local specialist availability and diagnostic infrastructure.

Another important aspect is population heterogeneity. Latin America is characterized by substantial diversity in ancestry, skin phototypes, socioeconomic conditions, occupational sun exposure, and access to healthcare services. This heterogeneity may influence lesion presentation, risk perception, healthcare-seeking behavior, and diagnostic pathways. However, many dermatological datasets used to develop artificial intelligence systems are not representative of the region and often incompletely report demographic variables relevant to fairness and generalizability. Recent studies evaluating dermatological artificial intelligence on diverse clinical images have shown that model performance may decrease when tested on more varied skin tones and less common diseases, reinforcing the need for validation beyond homogeneous benchmark datasets [19].

These factors create a regional gap between the clinical need to improve skin lesion assessment and the availability of representative data for algorithm development. For Latin America, the problem is not only the absence of high-performing models, but also the lack of locally validated systems capable of operating reliably under conditions of data scarcity, population diversity, and variable access to specialists. A model that achieves good global accuracy but fails to identify minority, premalignant, or malignant categories would have limited clinical value and could increase risk if used in triage scenarios without adequate safeguards.

Therefore, the problem addressed in this study is the need to improve the automated characterization of pigmented facial lesions in a context where the true disease burden may be incompletely measured, local datasets are limited, and diagnostic resources are unevenly distributed. From a Latin American perspective, artificial intelligence should be considered as a potential tool for clinical decision support, prioritization, and referral assistance, rather than as an autonomous replacement for medical diagnosis. This framing requires an evaluation that goes beyond global accuracy and includes class-balanced metrics, sensitivity in clinically relevant categories, and probability calibration.

In summary, pigmented facial lesions constitute a relevant clinical and public-health problem in Latin America due to solar exposure, population heterogeneity, diagnostic overlap among lesion types, incomplete epidemiological surveillance, and unequal access to dermatological specialists. These factors justify the development of robust, vision-only computational approaches—such as transfer-learning via foundation models—capable of supporting early identification and prioritization of lesions requiring specialist evaluation.

3. Method

3.1. Study Design

This study follows a computational, retrospective, and experimental design focused on the automated classification of facial pigmented skin lesions. The work is framed as a multi-source machine learning study, where dermoscopic images are used to train and evaluate a vision-only classification model for seven diagnostic categories. The study does not involve direct patient recruitment or prospective clinical intervention. Instead, it uses publicly available dermatological datasets and the MEDICON 2026 challenge dataset to investigate whether multi-source dermatology pretraining can improve classification performance under limited target-domain data.

3.2. Data Sources

The target MEDICON 2026 dataset initially provided 777 original images of facial pigmented lesions, including seven diagnostic categories:

1. Atypical nevus
2. Lentigo maligna
3. Lentigo maligna melanoma
4. Pigmented actinic keratosis
5. Seborrheic keratosis
6. Seborrheic-lichenoid keratosis
7. Solar lentigo

To mitigate target data limitations and handle substantial class imbalances, a multi-source data harvesting pipeline was implemented incorporating HAM10000, PAD-UFES-20, ISIC 2019, DERM12345, BCN20000, and ISIC_ARCHIVE. Datasets and samples were included according to the following criteria.

3.3. Inclusion Criteria

Datasets and samples were included according to the following criteria:

- Publicly available dermatology datasets with skin lesion images.
- Images associated with diagnostic labels or clinically meaningful categories.
- Datasets containing pigmented, premalignant, or malignant skin lesions relevant to the target task.
- Samples with sufficient image quality for model training.
- MEDICON samples containing valid dermoscopic images with resolvable diagnostic labels.
- External datasets suitable for label harmonization and training set expansion.

3.4. Exclusion Criteria

Samples were excluded when they met one or more of the following conditions:

- Missing or corrupted image files.
- Images with unreadable format or severe acquisition artifacts.
- Samples without usable diagnostic information.
- Duplicate images detected across datasets or within the same dataset.
- Labels that could not be mapped or interpreted for auxiliary training.
- Non-dermatological images or images unrelated to skin lesion classification.
- Metadata records with inconsistent or invalid values that could not be corrected.

3.5. Because the external datasets differed in original taxonomy, label harmonization was performed using dataset APIs to map external diagnostic labels directly to the target seven-class MEDICON taxonomy.

Any external samples whose clinical labels could not be directly mapped to one of these seven target categories were excluded to maintain a homogenous training pool.

3.6. Procedures

3.6.1. Problem Identification

The study addresses the need for improved automated characterization of pigmented facial lesions in Latin American-relevant contexts, where access to dermatology specialists may be uneven and locally representative datasets remain limited. The central computational problem is to classify visually similar facial pigmented lesions while maintaining class-balanced performance and calibrated probabilistic outputs.

The problem was formulated as a seven-class supervised classification task using dermoscopic images. Special attention was given to class imbalance, visual similarity among lesion types, and the risk of overfitting due to the small size of the target dataset.

3.6.2. Data Collection and Information Gathering

Dermatological data were gathered from the MEDICON 2026 challenge dataset and five external public datasets. The external datasets were reviewed according to modality, diagnostic taxonomy, number of samples, availability of metadata, and clinical relevance to pigmented lesion classification.

The purpose of incorporating external datasets was precisely to map and merge all applicable labels into a single homogeneous classification system. By doing so, the harmonized dataset exposed the model to a broader range of dermatological patterns relevant to the target taxonomy, directly addressing severe class imbalances.

3.6.3. Data Preprocessing

All images were standardized before model training and feature extraction. The preprocessing pipeline primarily consisted of converting various image formats (PNG, BMP, and TIF) to high-quality JPGs to ensure uniform compatibility with the

downstream feature extractor. Furthermore, duplicate images across the merged sources were programmatically removed to prevent data leakage. Clinical metadata was used exclusively during this stage to isolate the correct target labels and clean the dataset. No synthetic visual data augmentation (such as geometric transformations or color perturbations) was applied to the dataset.

3.6.4. Feature Extraction

Due to the limited original target data and the computational overhead of training deep neural networks from scratch, this study adopted a transfer-learning approach via a foundation model. Instead of attempting multi-source pretraining, a frozen, pre-trained Google Derm Foundation model was utilized as a static feature extractor. Standardized images were passed directly into the foundation model to extract rich, 6144-dimensional embeddings for all samples. This unimodal methodology captures complex dermatological patterns leveraging the foundation model's extensive prior knowledge without actively updating its underlying weights.

3.6.5. Model Development

The classification architecture is a strictly unimodal, vision-only pipeline. Clinical metadata (such as patient demographics or anatomical sites) was utilized solely for dataset organization and completely excluded from the predictive classifier.

The predictive network built on top of the frozen foundation model's 6144-dimensional embeddings consists of a lightweight Multi-Layer Perceptron (MLP). The MLP features two dense layers comprising 256 and 128 units, culminating in a seven-class probabilistic softmax output. Rather than handling class imbalance via metadata feature fusion, explicit class weighting was applied directly to the harmonized 11,020-sample training set during the MLP's training phase.

3.6.6. Model Training

The model was trained using supervised learning. The target objective was to minimize classification loss while improving

class-balanced performance. Due to the expected imbalance among diagnostic classes, strategies such as class weighting, balanced sampling, focal loss, or label smoothing may be considered.

The training process included:

- Extracting fixed embeddings for all harmonized images using the pre-trained Google Derm Foundation model.
- Training the Multi-Layer Perceptron (MLP) directly on the fully merged 11,020-sample dataset.
- Applying explicit class weights during training to handle dataset imbalance.
- Validating using strictly isolated 350-sample stratified splits from the original MEDICON dataset.
- Selection of the best-performing model according to validation performance.

Model training emphasized not only overall accuracy, but also sensitivity to minority and clinically relevant classes.

3.6.7.

Pilot Evaluation

A preliminary evaluation was performed using the 350-sample validation data to estimate model performance before final testing. The validation dataset was composed exclusively of the Medicon Challenge Dataset images. The model achieved a Macro-AUC of 0.6838, indicating a moderate baseline ability to discriminate and rank the seven diagnostic categories. The overall Accuracy reached 0.3886, alongside a highly calibrated Brier Score of 0.1079. The preliminary model achieved the following performance values:

Metric	Value
Macro-AUC	0.6838
Average Recall	0.2904
Macro-F1	0.3012
Brier Score	0.1079
Accuracy	0.3886

Metric	Value
Final Project Score	0.5041

3.7. While the Brier Score indicates strong probabilistic calibration, the model yielded an Average Recall of 0.2904 and a Macro-F1 of 0.3012. This discrepancy suggests that while the foundation model successfully generates calibrated probabilities, achieving high class-level sensitivity across overlapping and visually similar minority classes remains a distinct challenge without further optimization.

3.8. Functional System

The functional system consists of a complete inference pipeline designed to receive dermoscopic images as input and produce class probabilities as output.

The system includes the following components:

- Image preprocessing module (format standardization).
- Feature extraction module (Google Derm Foundation model embeddings).
- Vision-only predictive MLP classifier.
- Prediction generation and output formatting module.

For each case, the system produces:

- The lesion identifier.
- The probability for each of the seven diagnostic classes.
- The final predicted class.

The system is intended as a research prototype for decision-support analysis, not as an autonomous diagnostic tool. Its potential clinical role would be limited to prioritization, referral support, or triage under medical supervision.

3.9. Evaluation Metrics

The model was evaluated using complementary metrics that capture different aspects of performance. Since medical classification tasks may be affected by class imbalance, the evaluation did not rely only on global accuracy.

3.9.1. Discrimination Metrics

Macro-AUC was used to evaluate the model's ability to distinguish each class from the others independently of the final classification threshold. This metric is especially useful when the model produces probabilistic outputs and when class imbalance is present.

A higher Macro-AUC indicates that the model can rank positive cases above negative cases across classes. However, Macro-AUC alone does not guarantee reliable class-level predictions.

3.9.2. Class-balanced Performance Metrics

Average Recall and Macro-F1 were used to evaluate whether the model performs consistently across all diagnostic categories.

Average Recall measures the mean sensitivity across classes and is important in clinical contexts where missing premalignant or malignant categories may have consequences. Macro-F1 summarizes the balance between precision and recall for each class and then averages the result across all classes.

These metrics are particularly relevant because a model may achieve acceptable accuracy while still performing poorly on minority classes.

3.9.3. Calibration and Global Performance Metrics

The Brier Score was used to assess the quality of the predicted probabilities. Lower Brier Score values indicate better probabilistic calibration. Calibration is important in medical decision-support systems because clinicians may use probability estimates to prioritize cases or decide whether additional evaluation is needed.

Accuracy was also reported as a global performance measure. However, it was interpreted cautiously due to the possibility of class imbalance.

The final project score combined multiple evaluation dimensions, including discrimination, class-balanced performance, calibration, and accuracy.

4. Experimental Setup

4.1. Dataset Composition

The experimental setup included one target dataset and five external datasets. MEDICON 2026 was used as the target domain, while HAM10000, PAD-UFES-20, ISIC 2019, DERM12345, and BCN20000 were used as auxiliary sources.

4.2. The target dataset provided dermoscopic images corresponding to seven facial pigmented lesion categories. External datasets contributed immense dermatological variability and, rather than being used for isolated representation learning, were mapped directly to the target taxonomy to expand and construct a unified 11,020-sample training pool.

5. Experimental Process

5.1. Data Preparation

The first stage involved organizing all datasets, standardizing image formats, and harmonizing diagnostic labels across all sources. The target MEDICON labels were preserved as the final seven-class classification taxonomy.

5.2. External dataset diagnoses were mapped directly to the target taxonomy and merged into a single, homogeneous training pool to directly address target-task class imbalances.

5.3. Quantitative Analysis

In addition to global performance metrics, a detailed class-level analysis was conducted to evaluate the model's behavior across the seven distinct facial pigmented lesion categories. A confusion matrix (Figure 1) was generated to identify specific misclassification patterns among visually overlapping entities (e.g., distinguishing benign solar lentigo from lentigo maligna).

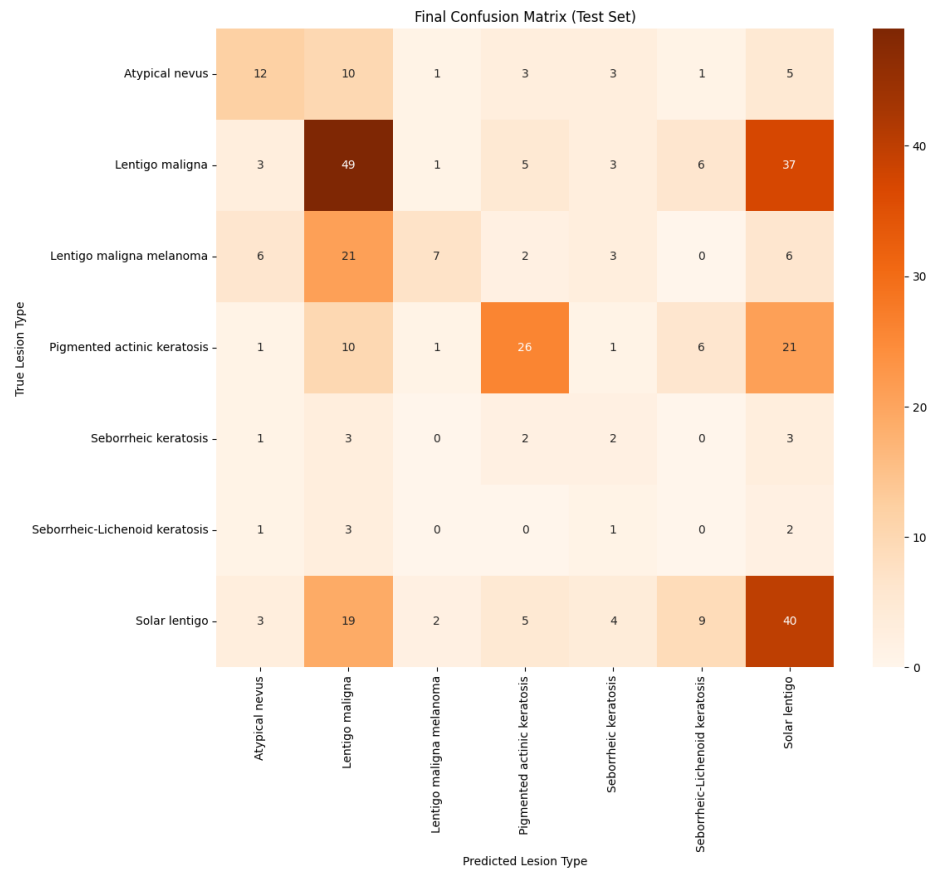


Figure 1. Final Confusion Matrix

Furthermore, per-class precision, recall, and F1-scores were calculated to determine if the foundation model disproportionately underperformed on rare or premalignant classes despite the dataset expansion. These results are presented in Figure 2.

	precision	recall	f1-score	support
Atypical nevus	0.44	0.34	0.39	35
Lentigo maligna	0.43	0.47	0.45	104
Lentigo maligna melanoma	0.58	0.16	0.25	45
Pigmented actinic keratosis	0.60	0.39	0.48	66
Seborrheic keratosis	0.12	0.18	0.14	11
Seborrheic-Lichenoid keratosis	0.00	0.00	0.00	7
Solar lentigo	0.35	0.49	0.41	82
accuracy			0.39	350
macro avg	0.36	0.29	0.30	350
weighted avg	0.45	0.39	0.39	350

Figure 2. Per-class metrics on the test set

5.4. Integrated Model Implementation

The final implementation utilized a strictly unimodal, vision-only classification pipeline built upon the extracted foundation model embeddings. The model output was exported as a prediction table containing the lesion identifier, class probabilities, and predicted class. This ensured reproducibility and compatibility with external evaluation.

6. Preliminary Analysis

The preliminary results indicate that the foundation model approach achieved moderate discriminative capacity but limited class-balanced classification performance. The Macro-AUC of 0.6838 suggests that the embeddings successfully captured useful visual patterns. However, the Average Recall of 0.2904 and Macro-F1 of 0.3012 reveal that the model still struggles to correctly identify several highly overlapping classes despite the massive dataset expansion. This discrepancy suggests that the model is better at ranking diagnostic probabilities than at producing reliable class-level predictions. In medical AI contexts, this distinction is important because a model may show acceptable discrimination while still failing to detect minority or clinically relevant lesion categories.

In contrast to the class-level sensitivity challenges, the model achieved a highly calibrated Brier Score of 0.1079. This indicates that the foundation model generates relatively reliable probabilistic confidence levels. Calibration is

especially relevant for decision-support tools, where accurately calibrated probabilities may safely influence referral prioritization or clinical follow-up, even if absolute classification accuracy remains moderate. From a Latin American perspective, these preliminary findings reinforce the need for cautious deployment. The model should not be interpreted as an autonomous diagnostic system. Instead, its current performance supports its use as an experimental decision-support prototype that may help explore triage, prioritization, or second-opinion workflows under medical supervision.

Future improvements should focus on increasing class-balanced performance through advanced optimization of the foundation model embeddings, improved class weighting strategies for heterogeneous lesions, and continuous validation on locally representative Latin American datasets.

7. Discussion

This study explored a vision-only transfer-learning strategy for the classification of facial pigmented lesions utilizing the pre-trained Google Derm Foundation model. The proposed framework was motivated by the highly limited size of the original target MEDICON 2026 dataset and the clinical complexity of differentiating benign, premalignant, and malignant pigmented facial lesions. In this context, external dermatology datasets were harvested and their diagnostic labels were mapped directly to the target taxonomy, creating a robust, unified 11,020-sample training pool to explicitly address severe data scarcity and class imbalances.

The preliminary results showed moderate discriminative capacity, with a Macro-AUC of 0.6838. This finding suggests that the foundation model's 6144-dimensional embeddings successfully captured relevant visual patterns, including pigmentation distribution, texture, border characteristics, and lesion morphology. However, the Average Recall of 0.2904 and Macro-F1 of 0.3012 indicate that the predictive classifier still has limited ability to provide consistent, class-level sensitivity across all diagnostic categories. This discrepancy is clinically important because a model may rank probabilities reasonably well while still failing to correctly identify underrepresented or visually similar minority classes.

The relatively low class-balanced performance may be explained by several factors. First, although the training pool was successfully expanded to 11,020 samples, the original target distribution was highly skewed, making it difficult

to fully balance the complex minority classes. Second, facial pigmented lesions frequently present heavily overlapping dermoscopic features, particularly among lentigo maligna, lentigo maligna melanoma, solar lentigo, pigmented actinic keratosis, and seborrheic keratosis. Third, differences in image acquisition, labeling criteria, and dataset origin between the incorporated external datasets and the target MEDICON task inevitably introduce domain shift, which the frozen foundation embeddings alone cannot fully mitigate.

In contrast to the lower recall, the highly favorable Brier Score of 0.1079 indicates strong probability calibration. In medical decision-support systems, calibrated probabilities are essential because clinicians may use model outputs to prioritize referrals or determine whether additional evaluation is required. The model's ability to produce reliable probabilistic confidence levels—even when its definitive top-1 prediction is incorrect—provides a safer foundation for clinical triage compared to an uncalibrated, overconfident system.

From a Latin American perspective, these findings reinforce the need for cautious interpretation. Vision-only artificial intelligence systems for dermatology should not be considered autonomous diagnostic tools, especially in regions where epidemiological surveillance, specialist availability, and access to dermoscopy may vary substantially. Instead, the proposed model should be understood as an experimental decision-support prototype with potential applications in triage, prioritization, or second-opinion workflows under medical supervision.

A relevant contribution of this work is its emphasis on class-balanced and calibration-aware evaluation. Accuracy alone would be insufficient in this setting because it may obscure poor performance in minority or clinically relevant categories. Therefore, metrics such as Average Recall, Macro-F1, Macro-AUC, and Brier Score provide a more comprehensive assessment of model behavior under data-scarce and clinically heterogeneous conditions.

Several limitations should be acknowledged. The current analysis was based on preliminary validation rather than external prospective testing. While the harmonized training dataset was large, the external datasets used to expand

it differed slightly in original taxonomy, population characteristics, and acquisition protocols. In addition, clinical metadata and demographic variables—such as patient age, anatomical site, or skin phototype—were entirely excluded from the predictive pipeline, limiting conclusions regarding demographic fairness and sub-population generalizability.

Future work should focus on advanced optimization of the foundation model embeddings, dynamic handling of domain shift, and validating the model on independent Latin American cohorts. Additional strategies may include end-to-end fine-tuning of the foundation model rather than keeping it frozen, test-time augmentation, uncertainty estimation, and expert-guided error analysis. Ultimately, clinical translation will require prospective validation, transparent reporting, and evaluation of model performance across diverse skin phototypes and healthcare settings.

8. Conclusions

This study proposed a vision-only transfer-learning framework utilizing the Google Derm Foundation model for the seven-class classification of facial pigmented lesions. The approach addressed a clinically relevant problem in dermatology, particularly in data-limited Latin American contexts where access to specialist evaluation, dermoscopy, and locally representative datasets may be uneven.

The preliminary results demonstrated moderate discriminative capacity, reflected by a Macro-AUC of 0.6838. However, the lower Average Recall and Macro-F1 values indicated limited class-balanced performance, suggesting that the model still struggles to consistently identify visually similar or highly imbalanced diagnostic categories. Conversely, the exceptional Brier Score of 0.1079 demonstrated highly reliable probabilistic calibration, a critical feature for safe clinical implementation.

Overall, the findings suggest that combining massive, multi-source dataset harmonization with foundation model embeddings can successfully mitigate target-domain data scarcity without requiring computationally expensive pretraining. Nevertheless, the current model should be interpreted only as an experimental decision-support prototype, not as an autonomous diagnostic system. Future research should prioritize external validation, optimized embedding adaptation, and evaluation in diverse Latin American populations to ensure clinical reliability, fairness, and generalizability.

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